

Numerical methods and deep learning

Phase-field fracture and differentiable simulation

Presented by
Sathiskumar A. Ponnusami

Queen Mary University of London · CEMS-Lab
Prepared for the UKACM Autumn School 2026

What learning can contribute

Approximation

Represent a complex relation between inputs and physical responses.

Repeated queries

Amortise a training investment across many suitable predictions.

A learned component

Propose a material response or a correction inside a numerical method.

Compare data generation, training, prediction and correction costs separately.

What a prediction must satisfy

Physical conditions

Boundary conditions, balance laws and admissible material states.

Numerical accuracy

Discretisation error, solver tolerance and the chosen observation.

Use beyond training

Unseen loads, geometries and material parameters require evaluation.

Assess prediction accuracy and physical consistency separately.

Coupling a network and a solver

Prediction with a learned component



Training through the numerical response



Backward sensitivity: loss → solve → model weights

Blue: forward evaluation. Orange: backward sensitivity. Each interface has a contract.

The six-hour learning sequence

Lecture questions

1. Fracture and numerical solution
2. Differentiation through a solve
3. Learning inside numerical methods

Practical evidence

1. A small PhAST forward calculation
2. A checked sensitivity and recovery
3. Train, reload and assess a proposal

Three lecture hours, followed by three practical hours.

A crack represented by a damage field

$d = 0$: intact

$d = 1$: fully damaged

$$\Pi(u, d) = \int_{\Omega} g(d) \psi_0(\varepsilon(u)) \, dx + G_c \Gamma_{\ell}(d) - W_{\text{ext}}(u)$$

$$\Gamma_{\ell}(d) = \frac{1}{2} \int_{\Omega} \left(\frac{d^2}{\ell} + \ell |\nabla d|^2 \right) dx$$

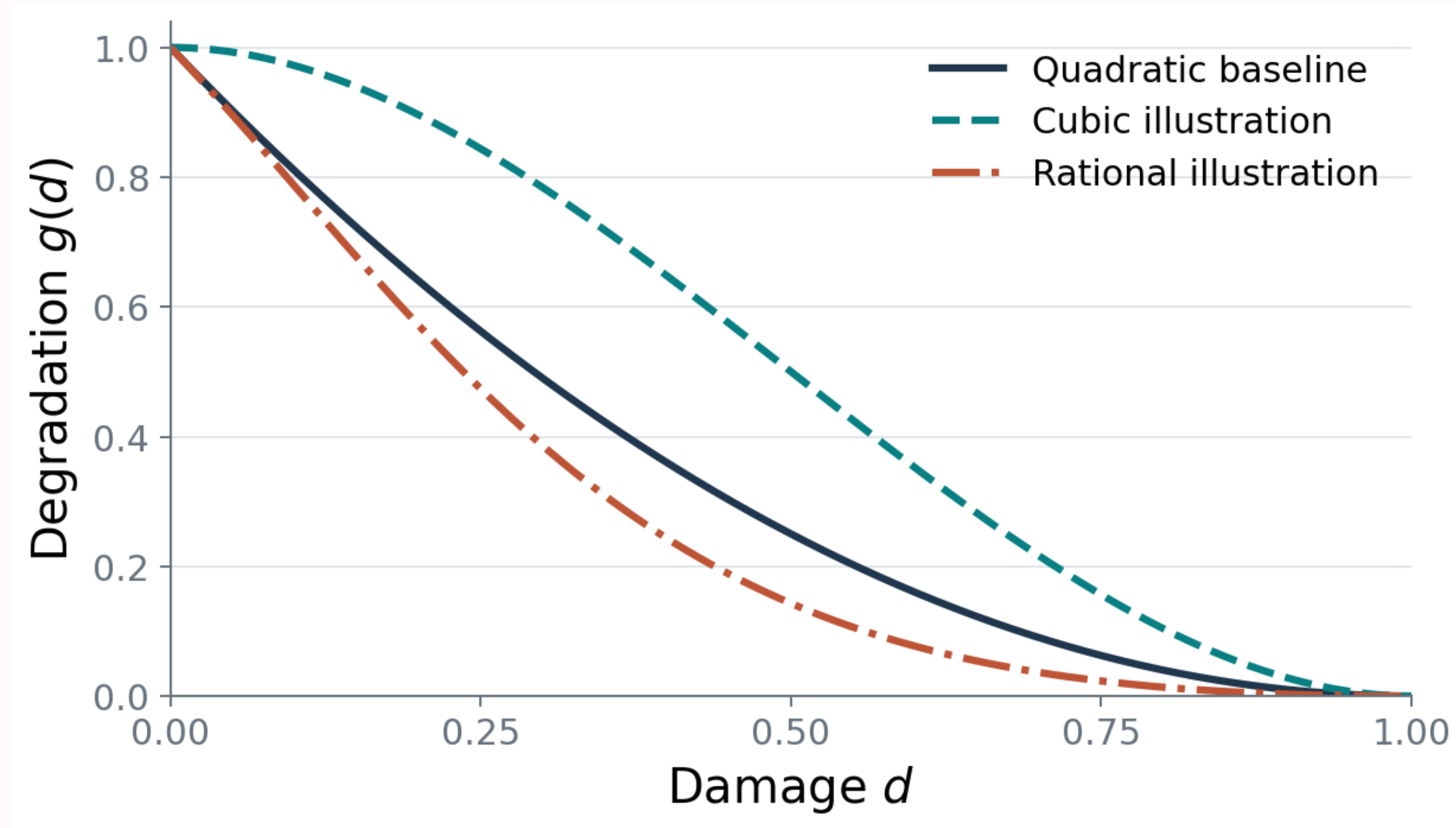
Stored elastic energy

Regularised crack cost

External work

Illustrative isotropic AT2 energy; tension/compression splits are a separate modelling choice.

How damage changes stiffness



$$g(d) = (1 - \eta)(1 - d)^2 + \eta$$

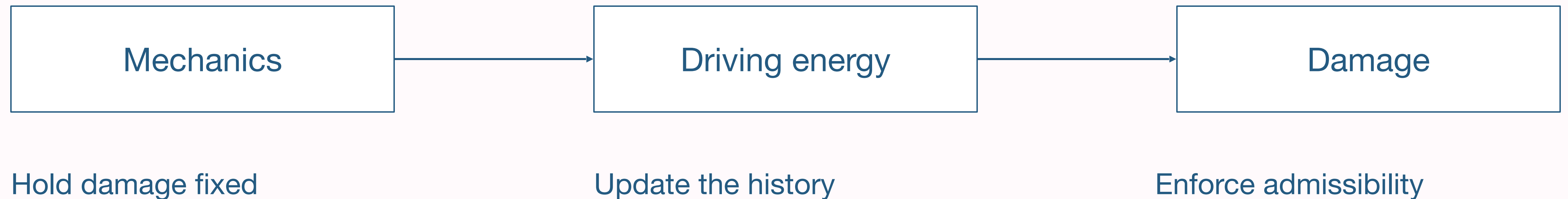
$$g'(d) = -2(1 - \eta)(1 - d)$$

Predict the sign of the derivative.
What changes near complete damage?

Quadratic baseline with illustrative cubic and rational alternatives.

The staggered solution

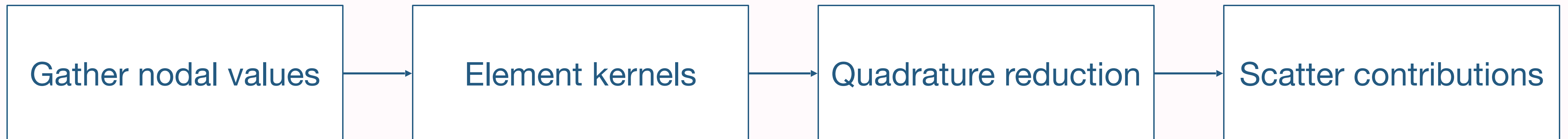
At a fixed load increment, update one coupled field at a time.



Repeat with the updated damage until the chosen convergence checks pass.

Load increment, staggered iteration and physical time are different indices.

Finite elements as tensor operations



Element, quadrature and component axes express the numerical structure.

The same operations can expose derivatives when the computational path supports them.

Differentiability depends on the operations along the computational path.

From an input to a final observation

$$z_{n+1} = S_n(z_n, p), \quad J = \frac{1}{2} \|Cz_N - y_\star\|^2$$



Input: a material parameter, a load parameter or model weights

Observation: a fixed map from the final state to the measured quantities

Differentiate the final scalar loss with respect to a defined input.

The backward pass through time steps

$$A_n = \frac{\partial S_n}{\partial z_n}, \quad B_n = \frac{\partial S_n}{\partial p}$$

$$\lambda_N = C^T (Cz_N - y_\star), \quad \lambda_n = A_n^T \lambda_{n+1}$$

$$\frac{dJ}{dp} = \sum_{n=0}^{N-1} B_n^T \lambda_{n+1}$$

Each use of the same parameter contributes to the final gradient.

Fixed initial state and observation map; parameter dependence enters through the updates.

Implicit differentiation of a solved equation

$$R(z, p) = 0, \quad R_z^T \lambda = \nabla_z J, \quad \nabla_p J_{\text{total}} = \nabla_p J - R_p^T \lambda$$

Forward calculation

Find a converged state satisfying the residual equation.

Backward calculation

Solve the transposed sensitivity system, then form the parameter derivative.

Requires a differentiable residual and a locally nonsingular state Jacobian.

Check the derivative before using it

$$\frac{dJ}{dp} \approx \frac{J(p+h) - J(p-h)}{2h}$$

Automatic differentiation

Differentiate the implemented computation.

Independent check

Use an analytical expression or a hand-written adjoint.

Finite differences

Compare several spacings at the same physical state.

A small time-stepping notebook

$$q_i^{n+1} = q_i^n + \frac{\alpha \Delta t}{\Delta x^2} (q_{i-1}^n - 2q_i^n + q_{i+1}^n)$$

q : scalar α : diffusivity $\Delta t, \Delta x$: fixed step sizes

Plot a forward trajectory.

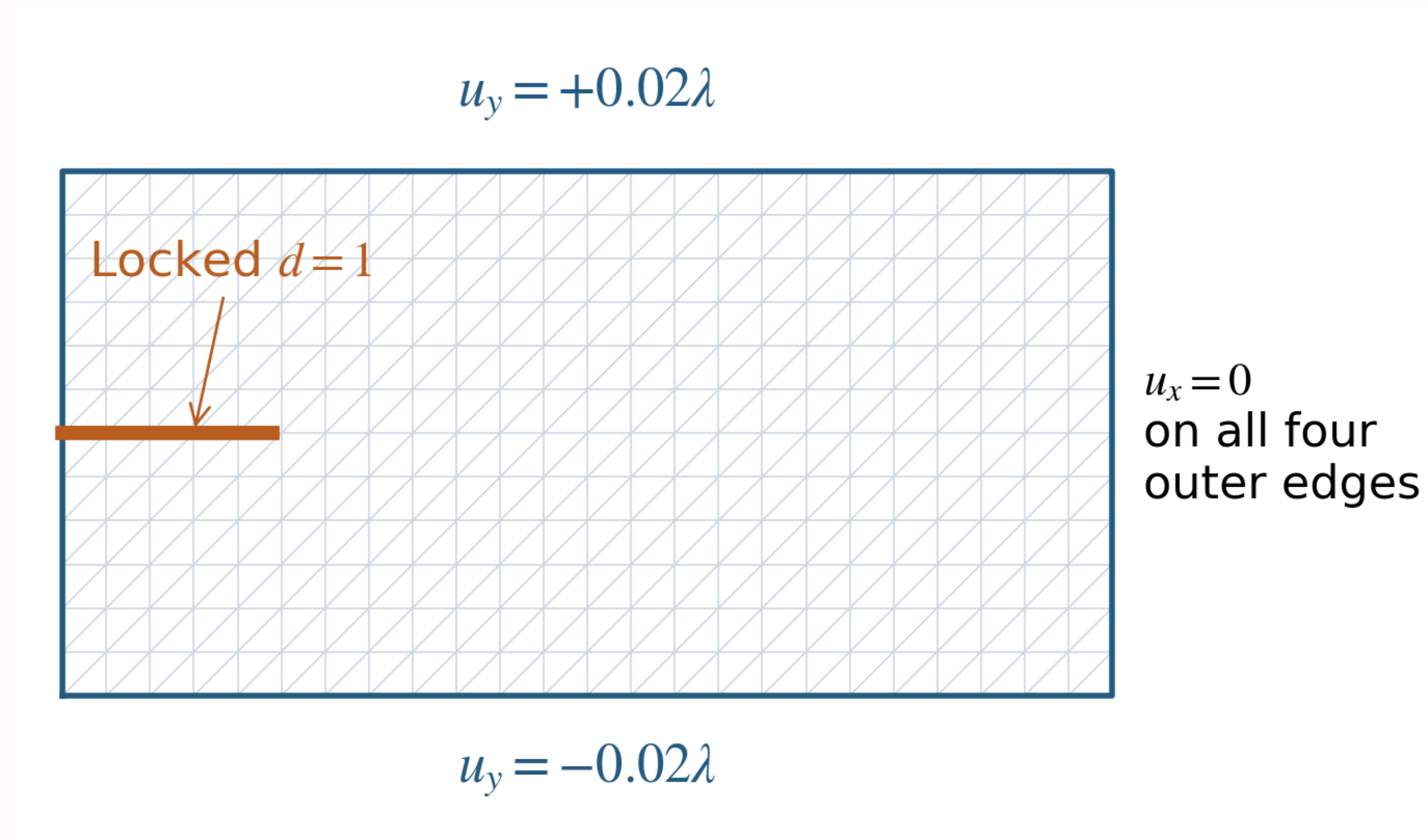
Differentiate a terminal observation.

Compare AD, manual reverse mode and finite differences.

Recover a synthetic parameter.

One-dimensional diffusion provides a compact example of time-step differentiation.

A mesh, a prescribed notch and boundary values

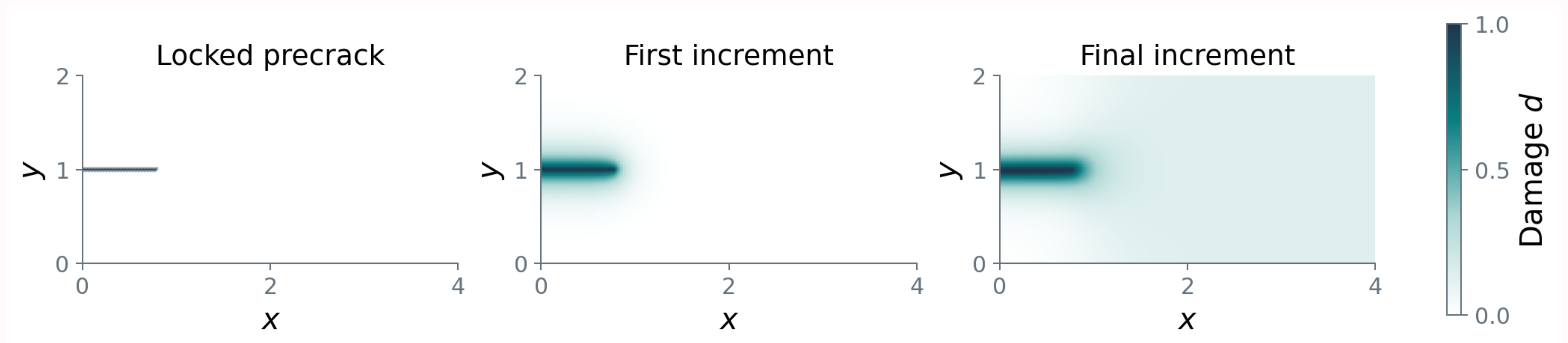


Represent the notch by prescribing fully damaged nodes on the intact mesh.

Inspect every boundary set and displacement component before solving.

Diagram uses a coarse mesh; the actual quick run has 8,385 nodes and 16,384 triangles.

Read the field as well as the response



Separate the prescribed seed, its diffuse profile and subsequent damage change.

Quasistatic AT2 response: initial damage, diffuse regularisation and loading-induced change.

Keep a reproducible numerical experiment

Input specification

Geometry, connectivity, materials, boundary values and loading.

Numerical specification

Algorithm, precision, tolerances and failure conditions.

Saved evidence

Field arrays, response history, configuration and source revision.

Reload the saved data into fresh variables, then reproduce a plot.

The practical learning cycle

Predict

What should change, and in which direction?

Compute

Run one short, inspectable calculation.

Explain

Use a field, a response and an independent check.

Companion book, executable notebooks and worked solutions: CEMS-Lab autumn-school.